# Yash Raj Shrivastava

Co-founder & Full-Stack Developer | B.Tech CSE (IILM, 2025–2029)
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## Summary

- Computer-science student combining software engineering with embedded systems and applied ML/GenAI.
- Built an open-source ESP32 quadruped robot and a ROS2 swarm-drone simulation; run production full-stack and local-LLM systems.
- Completed the 360 Labs AI test work week; Sagar Defence Engineering technical interview (drone/AI internship).

## Experience

**Co-founder & Product Designer — Stealth-mode SaaS** **(ongoing)**

- Leading product strategy, technical architecture, and investor outreach for a reservation and analytics SaaS platform.
- Owns the full product lifecycle: ideation, UX design, full-stack development, and deployment.

**Lead Developer — Gurtai Bot** **(ongoing)**

- Built a Python-based Telegram AI assistant with local LLM integration via Ollama for privacy-first, zero API-cost responses.
- Designed a plugin-based modular architecture for extensible workflow automation; containerized with Docker.

**SIH 2026 (ISRO) — PS171 Browser Agent** **Optimization & Testing Lead**

- PII-aware browser agent (WXT/React extension + backend) for ISRO's Smart India Hackathon; led E2E test automation, performance optimization, and live-run diagnostics.
- Produced judge-facing project documentation and demo assets for the final submission.

**360 Labs — AI Test Work Week** **(Aug 2026)**

- Completed an intensive 1-week AI development program: rapid decision-making, fast shipping, validated output under time pressure.

**Sagar Defence Engineering — Technical Interview (Drone/AI Intern)** **(Aug 2026)**

- Interviewed on robotics projects, embedded systems, and defence-sector requirements; assessed interview process for defence hiring.

## Selected Projects

**RiceScan — AI Crop-Disease Detection** **(Python, EfficientNetV2S, OpenCV)**

- Built an AI agricultural pipeline: image capture → segmentation → classification → quality-percentage estimation (EfficientNetV2S, custom camera + controlled lighting).
- Integrated a local knowledge base for symptoms and treatment; designed a user-feedback loop for model retraining.

**Gurtai — Local-LLM AI Assistant** **(Python, Ollama, Docker)**

- Architected a Telegram assistant running local LLMs via Ollama — privacy-first, zero API cost; plugin-based architecture for workflow automation.

**Sesame Robot — ESP32 8-DOF Quadruped (Open Source)**

- Built a mini quadruped from scratch: ESP32 DevKitC, 8× MG90S metal-gear servos, 3D-printed PLA chassis; custom firmware (Wi-Fi AP, captive-portal browser control, REST API, OLED face expressions).
- Pinned ESP32Servo v3.0.9, allocated PWM timers for all 8 servos; diagnosed a 4 A power failure via measurement and thermal modeling, fixed with an ECO firmware actuation limit (2 A verified).
- Walk gait achieved; Gazebo (ROS2) simulation in WSL2; code at github.com/Yashop965/sesame-robot ( INR 3,150 total cost).

**Swarm Drone System — ROS2 Humble** **(URDF, Gazebo, MQTT)**

- Designed a carrier hexacopter that carries 4–6 autonomous sub-drones with capability-aware task scoring and autonomous docking; full simulation workspace with 4/4 task assignments validated.

## Software Projects

**CamPass — Student Safety & Tracking** **(Flutter, Node.js, PostgreSQL, FCM)**

- Real-time geofence tracking, SOS alerts, and smart notifications via Firebase Cloud Messaging; digital pass system with barcode scanning and role-based access.

## Hardware & Robotics Projects

**Sesame Robot — ESP32 8-DOF Quadruped (Open Source)**

- Built a mini quadruped from scratch: ESP32 DevKitC, 8× MG90S metal-gear servos, 3D-printed PLA chassis; custom firmware (Wi-Fi AP, captive-portal browser control, REST API, OLED status).
- Pinned ESP32Servo v3.0.9, configured PWM timer allocation for 8 servos; 4 A power failure root-caused and fixed via firmware ECO mode + upgraded 5 A buck (verified 2 A).
- Basic walk gait achieved; Gazebo/ROS2 simulation in WSL2; open source at github.com/Yashop965/sesame-robot.

**Swarm Drone System — ROS2 Humble Simulation**

- Carrier hexacopter carrying 4–6 autonomous sub-drones; capability-aware task scoring, ROS→MQTT/TCP mesh bridge, Gazebo physics with 4/4 validated task assignments (clean colcon build, 1.29 s).
- Full stack: ground-station UI → mother drone → sub-drone mesh → return-and-dock cycle with failsafe navigation.

**Other Hardware — DIY Cameras & Drone Component Testing**

- Custom camera prototypes with controlled lighting and exposure-control systems; drone ESC (30 A, Hobbywing, DShot) and motor-driver testing for flight stability and power efficiency.

## Experiments & Ops

- **Ops:** Docker for reproducible environments (Gurtai, dev fleets); CI/CD failure triage (local-pass/CI-fail); E2E test automation (Playwright).
- **Local AI inference:** Ollama and LM Studio — running and evaluating local LLMs for agent experiments and Gurtai; prompt and context tuning.
- **Simulation:** ROS2 Humble + Gazebo headless testing in WSL2 — URDF modeling, gait-pattern validation (4/4 task assignments on the swarm system).
- **Lab work:** IoT & Robotics Foundation Lab (CSE25309) — ESP32/I2C/PWM servo control, circuit simulation (Tinkercad), 8-DOF quadruped project.

## Skills

- **Languages:** Python, JavaScript, TypeScript, C++, Dart, SQL, Bash
- **AI & ML:** GenAI applications, prompt engineering, LLMOps, AI agents & chatbots, local inference (Ollama, LM Studio), RAG concepts, OpenCV, EfficientNet (transfer learning)
- **Web & Mobile:** Next.js, React, Node.js, Express, Flutter, Prisma/PostgreSQL, Firebase, Docker
- **Embedded & Hardware:** ESP32, Raspberry Pi, Arduino, ROS2 (Humble), Gazebo, servo/PWM/I2C, breadboard & schematic-level circuit work
- **Tools:** Git, CI/CD debugging, Playwright E2E, Vercel, ESLint, WSL2, AWS (basics)

## Education

- **IILM University, Gurugram** — B.Tech, Computer Science & Engineering (2025–2029)

## Certifications

- DeepLearning.AI — *AI Python for Beginners*; *Build Rich-Context AI Apps with Anthropic*
- 360 Labs — AI Test Work Week (Aug 2026)